CHAPTER 2

# Limits of sequences

S14, S16, S19, S20

**Before you begin · Inequalities and geometric sequences**

Without a calculator, in six minutes. Answer briefly. Use calculation with fractions and powers. Then choose the targeted follow-up indicated under the questions.

1. Bound $\frac{1}{n+1}$ for $n \geq 1$.
2. Compare $0.7^{n+1}$ and $0.7^n$ for $n \geq 0$.
3. Give the sign of $\sqrt{n^2+1} - n$.
4. Rationalise $\sqrt{x} - 1$.
5. Is an increasing sequence necessarily bounded above?
6. Give an example of a bounded non-constant sequence.

**Answers**

$0 < \frac{1}{n+1} \leq \frac{1}{2}$ ; $0{,}7^{n+1} < 0{,}7^n$ ; positive; multiply by $\sqrt{x}+1$; no; for example $(-1)^n$.

**Targeted review after the diagnostic check**

Redo only the questions you got wrong. For each question: **1.** Take the reciprocals of two positive numbers; R16 recalls the effect of the sign. **2.** Write the next term as 0.7 times the previous; R02. **3.** Compare the positive squares before the roots. **4.** Multiply by the conjugate and expand its product. **5.** Try the sequence with general term n; distinguish variation and bound. **6.** Test the powers of -1 over six indices; R02. The R questions serve as review prompts with the teacher, even if their collective routine is planned for later.

**Objectives**

Distinguish convergence and divergence; calculate a limit; use comparison, the squeeze theorem and monotone convergence.

## Course notes

**Definitions**

$u_n \to \ell \in \mathbb{R}$ means that any open interval containing $\ell$ contains all terms from a certain index. We say that $(u_n)$ converges to $\ell$. $u_n \to +\infty$ means that for every real number $A$, there is an index $N$ such that $n \geq N \Rightarrow u_n \geq A$. $u_n \to -\infty$ means that for every real number $A$, there is $N$ such that $n \geq N \Rightarrow u_n \leq A$.

A sequence may have no limit, not even an infinite one: $(-1)^n$ oscillates between $-1$ and 1. Every convergent sequence is bounded, but the converse is false.

## Operations and comparisons

For two convergent sequences with limits $a, b$, the sum tends to $a+b$, the product tends to $ab$ and the quotient to $\frac{a}{b}$ if $b \neq 0$. Rules involving infinity are used while checking signs. The forms $+\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$ do not allow a direct conclusion.

**Example.** $\frac{3n^2-n+1}{2n^2+4} = \frac{3-\frac{1}{n}+\frac{1}{n^2}}{2+\frac{4}{n^2}} \to \frac{3}{2}$. For $\sqrt{n^2+n}-n$, multiplying by the conjugate gives $\frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}$.

**Theorems**

If $u_n \leq v_n$ from a certain index onwards and $u_n \to +\infty$, then $v_n \to +\infty$. If $a_n \leq u_n \leq b_n$ from a certain index and $a_n, b_n \to \ell$, then $u_n \to \ell$ (squeeze theorem). Every increasing sequence bounded above converges. Every decreasing sequence that is bounded below converges. This theorem is assumed.

**Proof by comparison.** For every $A$, the terms of $u$ exceed $A$ from some index onwards $N_1$. The inequality $v_n \geq u_n$ holds beyond some index $N_2$. For $n \geq \max(N_1, N_2)$, $v_n \geq A$.

**Proof: increasing and not bounded above.** For every $A$, the absence of an upper bound gives $N$ such that $u_N > A$. Since it is increasing, $n \geq N \Rightarrow u_n \geq u_N > A$, hence $u_n \to +\infty$. Not being bounded above is not enough on its own.

## Geometric sequences

$$q^n \longrightarrow \begin{cases} 0 & \text{if } -1 < q < 1, \\ 1 & \text{if } q = 1, \\ +\infty & \text{if } q > 1. \end{cases}$$

If $q \leq -1$, $(q^n)$ has no limit: alternation for $q = -1$; even and odd subsequences of opposite signs and with unbounded moduli for $q < -1$.

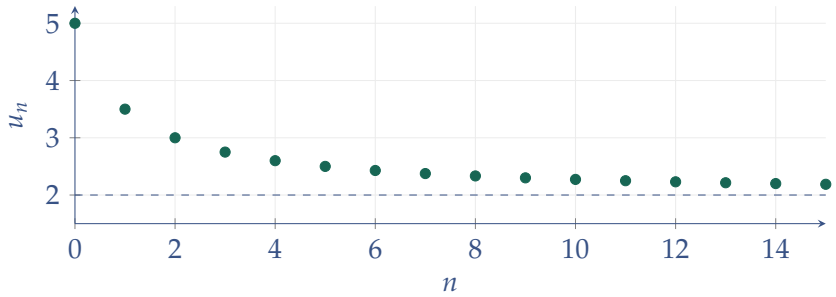
**Bernoulli and geometric limits.** Fix $a \geq 0$. Let us prove $P(n)$ : "$(1+a)^n \geq 1+na$". At index 0, both sides equal 1. Fix $n \geq 0$ and suppose $P(n)$ true. Since $1+a > 0$,

$$(1+a)^{n+1} \geq (1+na)(1+a) = 1+(n+1)a+na^2 \geq 1+(n+1)a.$$

This is $P(n+1)$ since $na^2 \geq 0$. By induction, the inequality holds for every $n \geq 0$. If $q > 1$, take $a = q-1 > 0$ and compare to $1+na$. If $0 < q < 1$, apply this result to $\frac{1}{q}$, then take reciprocals. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ and squeezing concludes. The case $q = 0$ is handled from index 1.

**Recursively defined sequence**

For $u_{n+1} = f(u_n)$, establish a bound and monotonicity to prove convergence. If $f$ is continuous at the limit $\ell$, taking the limit gives $\ell = f(\ell)$ (continuity discussed in Chapter 4). Solving this equation never proves, on its own, that the sequence converges.




The points get closer to 2 without reaching it: $u_n = 2 + \frac{3}{n+1}$.

## Give meaning to the limits

### 2.1 A band and all subsequent indices

First use: S14.

**Definition · Converging to $\ell$**

We say $(u_n)$ converges to $\ell$ if, for any precision $\varepsilon > 0$, there exists an index $N$ such that $n \geq N$ implies $|u_n - \ell| < \varepsilon$. The index depends on the precision, but once chosen, it works for all the following terms.

**Worked example · Calculating a valid index**

For $u_n = 2 + \frac{3}{n+1}$, $|u_n - 2| = \frac{3}{n+1}$. The precision 0,1 requires $n + 1 > 30$, so $n \geq 30$. Index 29 does not work: the gap there is exactly 0,1. For any precision, every $N$ such that $N + 1 > 3/\varepsilon$ works.

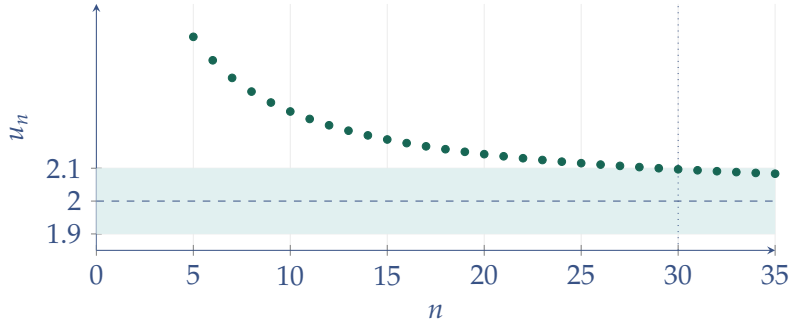



**Figure 2.1.** Band $]1{,}9;2{,}1[$ : all terms from index $n \geq 30$ remain there.

**A1 · Immediate application** For $v_n = 5 - \frac{2}{n+1}$, determine the first index from which $4{,}95 < v_n < 5.05$. Check the previous index and explain why all later indices work.

**A2 · Counterexample** "There is a term in each band around 0, so the sequence converges to 0." Refute with a sequence whose even terms are zero and whose odd terms are 1.

### 2.2 Transform before concluding

First use: S16 and S19.

**Worked example · A form does not determine a limit**

The three quotients $n/n$, $n/n^2$ & $n^2/n$ all have the form $\infty/\infty$. Yet their limits are 1, 0 and $+\infty$. For $\frac{3n^2-n+1}{2n^2+4}$, dividing numerator and denominator by $n^2$ gives the limit $\frac{3}{2}$. The transformation is valid for $n \geq 1$, which is enough for a limit.

**A3 · Choose before calculating** Determine the limits of $\frac{2n+3}{n^2+1}$, $\sqrt{n^2+n}-n$ and $\frac{(-1)^n}{n+1}$. Name, for each, the decisive transformation or theorem.

A first step: In the second expression, multiply by the conjugate. In the third, bound the absolute value.

### 2.3 Converge, then identify

First use: S20; identification by continuity after S32.

**Property · Monotone convergence · accepted without proof**

An increasing sequence that is bounded above converges to a real number. The theorem says neither that it reaches its limit, nor that its limit is any one of its upper bounds.

**Worked example · A complete chain**

Let $u_0 = 1$ and $u_{n+1} = \sqrt{2+u_n}$. The interval $[1;2]$ is stable because $1 \leq u_n \leq 2$ implies $\sqrt{3} \leq u_{n+1} \leq 2$. On this interval, $2 + x - x^2 = (2-x)(x+1) \geq 0$, therefore $u_{n+1} \geq u_n$ after comparison of positive squares. The sequence converges to $\ell \in [1;2]$. After continuity, $\ell = \sqrt{2+\ell}$ implies $(\ell-2)(\ell+1) = 0$; only $\ell = 2$ lies in $[1;2]$.

**A4 · Fix a line of reasoning** Seq. $a_0 = 1$, $a_{n+1} = 2a_n$ satisfies a relation whose only fixed point is 0. A student concludes $a_n \to 0$. Show the mistake and determine the actual behaviour.

**A5 · Independent proof** Let $b_0 = 0$, $b_{n+1} = \frac{1}{3}b_n + 2$. Prove $0 \leq b_n < 3$, then that it is increasing. Deduce convergence and determine the limit by the explicit formula or, after S32, by continuity.

### 2.4 Do not forget negative ratios

First use: S19 and S21.

**Property · Behaviour of $q^n$**

If $|q| < 1$, $q^n \to 0$. If $q > 1$, $q^n \to +\infty$. For $q = 1$, the sequence is constant. For $q \leq -1$, it has no limit, not even an infinite one: the signs alternate, and the two subsequences of even and odd indices have incompatible behaviours.

**Proof.** For $q > 1$, write $q = 1 + a$ with $a > 0$. Induction gives $(1+a)^n \geq 1 + na$, which tends to $+\infty$ ; comparison gives the result. For $0 < q < 1$, apply this result to $1/q$ and take the reciprocal. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ implies $q^n \to 0$ by the squeeze theorem. The case $q = 0$ is immediate from index 1. □

**Worked example · Common ratio and behaviour**

For $q = -2$, the even-index terms $4^k$ tend to $+\infty$ and the odd ones $-2 \times 4^k$ to $-\infty$ : no common limit. For $q = -1/2$, the signs also alternate, but the absolute value $2^{-n}$ tends to 0; the sequence converges.

**A6 · Compare properties** For $(-1)^n$, $-\frac{1}{n+1}$ & $n$, state: bounded? monotonic? convergent ? Give justifications, not just a table of answers.

**One minute, spoken aloud.** Explain why "increasing" means neither "positive" nor "tends to $+\infty$ ".

**Delayed review · S33 then S66.** Go back to identifying the limit by continuity; for a sequence of integrals, rebuild the same chain: bounds, monotonicity, convergence, identification.

## Investigation

**A limit is not a last term - 30 min.**

Consider the sequence $(u_n)$ defined on $\mathbb{N}$ by $u_n = 2 + \frac{3}{n+1}$.

1. Calculate $u_0$, $u_9$ and $u_{99}$. Is there an integer $n$ such that $u_n = 2$? Justify.

2. Determine the smallest integer $N$ such that, for every integer $n \geq N$, $|u_n - 2| < 0{,}01$.

3. Let $\varepsilon > 0$. Give, in terms of $\varepsilon$, an integer $N$ such that $|u_n - 2| < \varepsilon$ for all $n \geq N$.

4. On $\mathbb{N}$ define $(v_n)$ by $v_n = 2 + \frac{(-1)^n}{n+1}$. Prove that it converges to 2 and that it is not monotonic. What does this example show?

## Progressive exercises

**E1 · Direct application.** We define on $\mathbb{N}$ the sequences $(u_n)$ and $(v_n)$ by $u_n = \frac{4n-1}{n+2}$ & $v_n = n^2 - 7n$. Determine their limit, justifying the transformations used.

**E2 · Application and justification.** We define $u_n = \sqrt{n+1} - \sqrt{n}$ for all $n \in \mathbb{N}$. Using the conjugate expression, determine the limit of $(u_n)$.

**E3 · Method to organise.** We define $u_n = \frac{\cos n}{n}$ for every integer $n \geq 1$. Bound $u_n$, then deduce the limit of $(u_n)$.

**E4 · Choice of strategy.** Seq. $(u_n)$ is increasing and satisfies $u_n < 5$ for all $n \in \mathbb{N}$. Justify that it converges and state what can be said about its limit. Is it necessarily equal to 5? Justify.

**E5 · Synthesis or proof.** We define $u_n = 3 - 2(-0{,}6)^n$ for all $n \in \mathbb{N}$. Determine the limit of $(u_n)$. Is this sequence monotonic? Justify.

## Work from the textbook or a past paper

**Barbazo, p. 62, Exercises 17 and 24.** In S16: exercise 17, to resolve an indeterminate form by dividing by a power of $n$. At home after S20: exercise 24, to write up an application of the squeeze theorem.

## Homework

**DM2 - Approximating a root - 50 min.**

Consider $u_0 = 0$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2+u_n}$. Questions 1, 2 and 4 are to be done after S20; question 3 will be revisited after the lesson on continuity in S31.

1. Prove by induction that all terms are defined and that $0 \leq u_n \leq 2$ for every $n \in \mathbb{N}$.

2. Prove by induction that $u_{n+1} \geq u_n$ for all $n \in \mathbb{N}$. Deduce that $(u_n)$ converges.

3. Using the continuity of the function $x \mapsto \sqrt{2+x}$, determine the limit of $(u_n)$.

4. Prove that $0 \leq 2 - u_{n+1} \leq \frac{2-u_n}{2}$ for all $n \in \mathbb{N}$. Deduce an upper bound for $2 - u_n$ in function of $n$, then an integer $N$ such that $0 \leq 2 - u_n \leq 10^{-3}$ for all $n \geq N$.

## MCQ A · definitions and operations

After S16. MCQ B after S20; fixed point after S31.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

**1.** $u_n = 3 + \frac{2}{n+1}$. Its limit is:

A. 0 B. 2

C. 3 D. $+\infty$

**2.** $\frac{2n^2+1}{5n^2+n}$ tends to:

A. $\frac{2}{5}$ B. $\frac{5}{2}$

C. 0 D. $+\infty$

**3.** $u_n \to 2$ and $v_n \to 0$. What can be stated without any further assumption?

A. $u_n / v_n \to +\infty$ B. $u_n v_n \to 0$

C. $u_n / v_n \to 0$ D. $u_n + v_n \to 0$

**4.** For $u_n = 2 + \frac{3}{n+1}$, the first index from which $|u_n - 2| < 0.1$ is:

A. 28 B. 29

C. 31 D. 30

**5.** Seq. $(-1)^n$ is:

A. converging to 0 B. bounded and divergent

C. increasing D. diverging to $+\infty$

## MCQ B · choosing a theorem

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

**6.** We know $u_n$ is increasing and $u_n \leq 5$. Necessarily:

A. $u_n \to 5$ B. $u_n \geq 0$

C. $u_n \to \ell \leq 5$ D. $u_n$ constant

**7.** $u_n = \frac{\sin n}{n}$ for $n \geq 1$. The direct tool is:

A. monotone convergence

B. a fixed-point equation

C. the law of large numbers

D. the squeeze theorem

**8.** An increasing sequence that is not bounded above:

A. tends to $+\infty$

B. may tend to 0

C. is always geometric

D. oscillates between two values

**9.** $\sqrt{n^2+n} - n$ tends to:

A. $+\infty$

B. 1

C. $\frac{1}{2}$

D. 0

**10.** If $0 \leq u_n \leq (\frac{3}{4})^n$ for every index:

A. $u_n$ is geometric

B. $u_n \to 0$

C. $u_n$ is decreasing

D. $u_n \to \frac{3}{4}$

**True or false · justification expected.** For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every convergent sequence is monotonic.

2. Every increasing bounded sequence converges.

3. If $u_n \to \ell$ & $u_n < 5$, then $\ell < 5$.

## MCQ solutions

1. **C.** The quotient tends to 0, so the sum tends to 3. Neither the numerator nor the first term gives the limit.

2. **A.** Divide numerator and denominator by $n^2$. The dominant terms are of the same degree.

3. **B.** The product tends to 2 $\times 0 = 0$. The quotient may be undefined, or change sign; the sum tends to 2.

4. **D.** $3/(n+1) < 1/10$ is equivalent to $n > 29$. At index 29, the gap is exactly 0.1.

5. **B.** The even terms equal 1, the odd ones $-1$; they cannot stay in the same small band around a real number. The sequence is bounded.

6. **C.** Monotone convergence. The constant sequences equal to $-1$ refute positivity and the limit 5; $5 - 1/(n+1)$ disproves constancy.

7. **D.** $-1/n \leq u_n \leq 1/n$, and both bounds tend to 0; no monotonicity is required.

8. **A.** For every threshold, a term exceeds it, as do all later terms since the sequence is increasing. This is the definition of the limit $+\infty$.

9. **C.** Using the conjugate, the expression is $1/(\sqrt{1+1/n}+1)$ for $n > 0$, hence $1/2$. Subtraction of infinities did not allow a conclusion.

10. **B.** The upper bound tends to 0. A sequence between the bounds can occasionally increase, for example by cancelling the even terms.

**True/false.** False: $(-1)^n/(n+1) \to 0$ is not monotonic. True by monotone convergence. False: $5 - 1/(n+1) \to 5$; only the weak inequality passes to the limit.

## Solutions to the applications and guidance

**Application 1.** The difference from 5 is $2/(n+1)$ ; we need $n+1>40$, so $n \geq 40$. At index 39, $v_{39}=4{,}95$ is excluded. The gaps decrease, hence validity for all subsequent indices.

**Application 2.** The sequence has terms equal to 0 in every band, but for $\varepsilon = 1/2$, every index is followed by an odd index whose term is 1. No $N$ guarantees the universal condition.

**Application 3.** The first quotient tends to 0 on dividing by $n^2$. The second expression is $\frac{n}{\sqrt{n^2+n}+n} = \frac{1}{\sqrt{1+1/n}+1} \to 1/2$. The third has absolute value $1/(n+1) \to 0$, so it tends to 0.

**Application 4.** $a_n = 2^n \to +\infty$. The equation $\ell = 2\ell$ is a necessary condition if a finite limit exists; it does not prove this existence.

**Application 5.** Initially $b_0 = 0$. If $0 \leq b_n < 3$, then $2 \leq b_{n+1} < 3$, so the invariant is passed on. $b_{n+1} - b_n = \frac{2}{3}(3-b_n) > 0$. Monotone convergence; the formula $b_n = 3 - 3(1/3)^n$ gives the limit 3. By continuity, $\ell = \ell/3 + 2$ also gives 3 once convergence is established.

**Application 6.** $(-1)^n$ is bounded between $-1$ and 1, not monotonic and not convergent because its even and odd subsequences have different limits. $-1/(n+1)$ is increasing, bounded between $-1$ and 0, and converges to 0. The sequence $n$ is increasing, not bounded above and diverges to $+\infty$.

**Key points**

Have students colour the terms that stay outside the band, then express "stay" with quantifiers.
Before S32, give continuity as an accepted result in an identification exercise
or use an explicit formula. The proof of monotone convergence itself is still assumed;
students must be able to apply it reliably.

## Teaching notes and solutions

**Guidance for the teacher**

Written test 2 comes before S19 and S20: assess the definitions and operations already met, without require monotone convergence, Bernoulli or the squeeze theorem. Draw horizontal bands around the limit and have students look for an index from which all the terms stay inside it. Require the convergence before solving the fixed-point equation. Show $(-1)^n$ to separate "bounded " and "convergent".

### Solutions to the activity

1. $u_0 = 5$, $u_9 = 2.3$ and $u_{99} = 2.03$. Since $\frac{3}{n+1} > 0$ at every index, $u_n > 2$: no term equals 2.

2. $|u_n - 2| < 0{,}01 \iff \frac{3}{n+1} < 0{,}01 \iff n+1 > 300 \iff n \geq 300$. The first suitable index is so $N = 300$. At index 299, the gap is exactly 0.01 and the strict inequality is not satisfied.

3. For $\varepsilon > 0$, let us take $N = \lceil \frac{3}{\varepsilon} \rceil$, the smallest integer greater than or equal to $\frac{3}{\varepsilon}$. If $n \geq N$, then $n+1 > \frac{3}{\varepsilon}$, so $|u_n - 2| < \varepsilon$. This choice is valid, even when it is not minimal.

4. $|v_n - 2| = \frac{1}{n+1} \to 0$, therefore $v_n \to 2$. But $v_0 = 3$, $v_1 = \frac{3}{2}$ & $v_2 = \frac{7}{3}$ : we observe a decrease and then a increase. This convergent sequence is not monotonic.

**E1.** For $n \geq 1$, $\frac{4n-1}{n+2} = \frac{4-\frac{1}{n}}{1+\frac{2}{n}} \to 4$, since the denominator tends to 1 $\neq 0$. Similarly $n^2 - 7n = n^2(1 - \frac{7}{n})$ ; the first factor tends to $+\infty$ and the second to 1 $> 0$, so the product tends to $+\infty$.

**E2.** The conjugate gives $\sqrt{n+1} - \sqrt{n} = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. The denominator tends to $+\infty$, therefore the difference tends to 0.

**E3.** For $n \geq 1$, $-1 \leq \cos n \leq 1$ and $n > 0$, therefore $-\frac{1}{n} \leq u_n \leq \frac{1}{n}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.

**E4.** The sequence is increasing and bounded above by 5, so it converges. Taking the limit in $u_n \leq 5$, its limit satisfies $\ell \leq 5$. It is not necessarily equal to 5: $u_n = 2 - \frac{1}{n+1}$ is increasing, bounded above by 5 and converges to 2.

**E5.** Since $|-0{,}6| < 1$, $(-0{,}6)^n \to 0$, therefore $u_n \to 3$. However $u_0 = 1$, $u_1 = 4{,}2$ and $u_2 = 2{,}28$: the sequence goes up then down and is not monotonic.

### Solutions to homework DM2

1. Let $P(n)$ : "$u_n$ is defined and $0 \leq u_n \leq 2$". It is true at index 0 since $u_0 = 0$. Fix $n \geq 0$ and suppose $P(n)$ true. We have $2 \leq 2 + u_n \leq 4$, so $u_{n+1} = \sqrt{2+u_n}$ is defined and $\sqrt{2} \leq u_{n+1} \leq 2$. Thus $P(n+1)$ is true. By induction, all the terms belong to $[0, 2]$.

2. Let $R(n)$ : "$u_{n+1} \geq u_n$ ". For $n = 0$, $u_1 = \sqrt{2} \geq 0 = u_0$. Fix $n \geq 0$ and suppose $R(n)$ true. The increasing nature of $x \mapsto \sqrt{2+x}$ on $[0, 2]$ gives

$$u_{n+2} = \sqrt{2 + u_{n+1}} \geq \sqrt{2 + u_n} = u_{n+1}.$$

Hence $R(n+1)$ holds. By induction, $(u_n)$ is increasing. It is bounded above by 2, so it converges to a real number $\ell \in [0, 2]$.

3. The function $x \mapsto \sqrt{2+x}$ is continuous at $\ell$. Taking the limit, $\ell = \sqrt{2+\ell}$. Since $\ell \geq 0$, this is equivalent to $\ell^2 - \ell - 2 = 0$, i.e. $(\ell - 2)(\ell + 1) = 0$. The root $-1$ is excluded; therefore $\ell = 2$.

4. Using the conjugate and $u_n \leq 2$,

$$0 \leq 2 - u_{n+1} = \frac{2 - u_n}{2 + \sqrt{2 + u_n}} \leq \frac{2 - u_n}{2}.$$

Let $T(n)$ : "$0 \leq 2 - u_n \leq \frac{2}{2^n}$". At index 0, $2 - u_0 = 2$. If $T(n)$ is true for a fixed integer $n \geq 0$, then

$$0 \leq 2 - u_{n+1} \leq \frac{1}{2}\frac{2}{2^n} = \frac{2}{2^{n+1}}.$$

The base case and the inductive step prove $T(n)$ for every index. Since $\frac{2}{2^{11}} = \frac{1}{1024} < 10^{-3}$, $n = 11$ suffices. This is an index guaranteed by an upper bound, with no claim of minimality. This question may be dealt with before question 3: the number 2 already appears in the bounds.

## Solutions to the textbook or past-paper work

**Exercise 17.** The quotients are initially of indeterminate form. For $n \geq 1$,

$$u_n = \frac{2n - 4}{n^2 + 1} = \frac{2\left(1 - \frac{2}{n}\right)}{n\left(1 + \frac{1}{n^2}\right)} \longrightarrow 0.$$

Similarly,

$$v_n = \frac{2 - 5n}{4n + 7} = \frac{\frac{2}{n} - 5}{4 + \frac{7}{n}} \longrightarrow -\frac{5}{4}, \qquad w_n = n\frac{-1 - \frac{10}{n^2} + \frac{4}{n^3}}{2 + \frac{3}{n} + \frac{1}{n^2}} \longrightarrow -\infty.$$

The last quotient tends to $-\frac{1}{2}$, and the factor $n$ to $+\infty$.

**Exercise 24.** For every $n \geq 1$, $-1 \leq (-1)^n \leq 1$. Since $n^2 > 0$, we get $-\frac{1}{n^2} \leq u_n \leq \frac{1}{n^2}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.